

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CS/IT-3005-CBGS

B.E. III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Discrete Structure

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) If U is a universal set and its two subsets A and B , then prove that $(A \cup B)' = A' \cap B'$
b) Show that the set Q of rational numbers is countable.
2. a) Show that $\left[(p \wedge q) \Rightarrow p \right] \Rightarrow (q \wedge \sim q)$ is a contradiction.
b) Show that the language $L = \{a^m : m = i^2, i \geq 1\}$ is not a finite state
3. Define eulerian path and circuit of a graph with an example for each. State the necessary and sufficient conditions for existence of an Eulerian path in connected graph.
4. a) Prove :
 - i) $A \times (B \cap C) = (A \times B) \cap (A \times C)$
 - ii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$b) Explain Pigeonhole principle with an example.

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[2]

5. a) Test the validity of argument:
if it rains, Ram will be sick
it did not rain
∴ Ram was not sick
b) Explain universal and existential qualifiers with example
6. Define the following with examples:
i) Multigraph ii) Isomorphic graphs
iii) Eulerian graph
7. a) Prove that the set
 $G = \{ \dots -4m, -3m, -2m, -m, 0, m, 2m, 3m, 4m \dots \}$
of multiples of integers by a fixed integer m is a group
with respect to addition.
b) Discuss ring and field with example.
8. Write short notes on :
a) Hasse diagram
b) Lattices
c) Binomial theorem
d) Permutations
